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Antifragility in the synchronization of oscillators on networks with communities

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Abstract

Antifragility refers to the counterintuitive phenomenon whereby a system improves its global performance in response to damage. Although antifragile behavior has been reported in a variety of complex systems, the structural and dynamical mechanisms underlying this response remain poorly understood. In this work, we investigate antifragility in the synchronization dynamics of identical oscillators on networks, focusing on the effect of damage defined as the reduction in the weight of an edge connecting pairs of nodes in the network. Using a linearized description of the Kuramoto model with normalized couplings, we introduce a spectral measure of antifragility based on the response of the second eigenvalue of the normalized Laplacian. This framework enables an analytical and numerical characterization of how local reductions in coupling strength can accelerate global synchronization. Through first-order perturbation theory, we show that the sign and magnitude of the response are controlled by the structure of the slowest relaxation mode. In networks with modular organization, this mechanism leads to a systematic distinction between intra- and inter-community links: damage to intra-community edges is typically antifragile, whereas damage to inter-community edges is fragile. These predictions are validated by numerical simulations on deterministic, random, and real-world networks with community structure. Overall, our results establish a direct connection between spectral properties, modular topology, and antifragility in synchronization processes.

1. Introduction

Complex systems are composed of many interacting components whose collective behavior emerges from the interplay between dynamics and underlying structure [1, 2]. A defining feature of such systems is their sensitivity to perturbations: local failures or structural damage can propagate through the system and significantly alter its global performance [3–9]. Understanding how damage affects dynamical processes on networks is therefore a central problem in the study of complex systems, with direct implications for the identification of vulnerabilities and the design of resilient architectures [3, 10].

A substantial body of work has investigated the impact of damage on transport and diffusion processes, including the optimization of flow networks [11], strategies for network repair after failures or attacks [12–14], and the effects of structural degradation on diffusive dynamics [15–17]. Damage-induced changes have also been explored in the synchronization of oscillators [18] and in networks of interacting neuronal units subject to targeted attacks or failures [19–21]. These studies typically focus on robustness or fragility, that is, on the ability of a system to preserve its functionality or on the onset of performance degradation as damage accumulates.

Beyond robustness and fragility, a more counterintuitive response has been proposed in which damage can enhance, rather than degrade, system performance. This concept, termed antifragility, was introduced by Taleb [22] to describe systems that benefit from variability, stress, or damage. Antifragility has

since been investigated in a wide range of contexts, including financial markets [23], renewable energy systems [24], ecological networks [25], and other complex adaptive systems [26]. Related phenomena have also been reported in transportation networks, where the removal or blockage of specific roads can lead to improved global traffic flow [27]. Despite these observations, antifragility remains far less understood than robustness or fragility, and the structural and dynamical conditions under which it emerges are still an open problem [26].

Recent efforts have begun to address this gap by proposing explicit dynamical frameworks to characterize antifragile responses. Examples include Boolean network models [28, 29], strongly coupled Coulomb fluids [30], and random walk dynamics on modular networks, where damage can enhance exploration efficiency [31]. In the context of synchronization, numerical studies of the Kuramoto model have shown that weakening specific couplings may accelerate global phase coherence [32], suggesting a deep connection between antifragility and network structure, particularly in systems with pronounced heterogeneity or community organization.

In this work, we present an analytical and numerical study of antifragility in synchronization dynamics on networks. We consider identical Kuramoto oscillators with normalized couplings and introduce a quantitative measure of functionality based on synchronization times. By analyzing the linear response of this functionality to infinitesimal structural damage, which we define as a reduction in the weight of an edge connecting two nodes, we establish a direct connection between local edge perturbations and global dynamical performance. Using first-order perturbation theory together with a spectral characterization based on the normalized Laplacian, we show that antifragility naturally emerges in networks with pronounced modular organization, where the slowest relaxation mode is strongly shaped by community structure. Our analysis builds on the antifragile responses previously identified through extensive numerical integration of the Kuramoto dynamics in [32], where a quantitative framework was introduced to evaluate how reductions in individual link weights affect synchronization times and to classify links as antifragile, fragile, or neutral. Motivated by those findings, we develop here a spectral framework for the analysis of synchronization dynamics, which provides analytical insight into the origin of antifragility and enables the numerical study of substantially larger networks.

The paper is organized as follows. In section 2, we introduce the theoretical framework for the dynamics of identical Kuramoto oscillators on networks and define measures of functionality associated with global synchronization and its response to infinitesimal link perturbations. In section 3, we develop a perturbative analytical approach to characterize the emergence of antifragility induced by infinitesimal edge damage, and apply it to deterministic lollipop graphs and to random networks with two communities. In section 4, we present numerical results that validate and extend the analytical predictions, first for lollipop graphs and two-community networks, and then for networks with multiple communities generated using the Lancichinetti–Fortunato–Radicchi model, as well as for real-world networks with empirically observed modular structure. Finally, section 5 summarizes the main results and discusses their broader implications. Overall, our findings clarify how community organization enables the conversion of local damage into enhanced global synchronization, providing a general framework for understanding antifragile behavior in networked dynamical systems.

2. General theory

In this section, we establish the theoretical framework used to quantify the impact of infinitesimal structural damage on synchronization dynamics. We introduce the Kuramoto model of identical oscillators on weighted networks and define measures of global functionality associated with global synchronization. We then formalize how localized perturbations to individual links alter the coupling structure and define antifragility as the linear response of synchronization times to such perturbations. Finally, we present a linear and spectral formulation of the dynamics, which serves as the analytical basis for the perturbative results developed in the following sections.

2.1. Kuramoto model of identical oscillators

We consider N identical phase oscillators on a connected, undirected network with nodes $i = 1, \dots, N$. The network connectivity is encoded in a symmetric adjacency matrix $\mathbf{A}$, where $A_{ij} = 1$ if nodes i and j are connected and $A_{ij} = 0$ otherwise. In particular, $A_{ii} = 0$ to avoid self loops. Links are assigned non-negative weights through an $N \times N$ matrix $\mathbf{\Omega}$ with entries $\Omega_{ij} \geq 0$, and we use the directed notation $i \rightarrow j$ for couplings from i to j . From $\mathbf{\Omega}$ we define the normalized coupling matrix $\mathbf{W}(\mathbf{\Omega})$ with elements

$$w_{i \rightarrow j}(\mathbf{\Omega}) \equiv \frac{\Omega_{ij}}{\mathcal{S}_i}, \quad (1)$$

where $\mathcal{S}_i = \sum_{l=1}^N \Omega_{il}$ is the generalized strength of node i [33]. By construction, $0 \leq w_{i \rightarrow j}(\Omega) \leq 1$ and $\sum_{l=1}^N w_{i \rightarrow l}(\Omega) = 1$. Then, the dynamics of identical Kuramoto oscillators on the weighted network is given by

$$\frac{d\theta_i(t)}{dt} = \sum_{j=1}^N w_{i \rightarrow j}(\Omega) \sin[\theta_j(t) - \theta_i(t)], \quad (2)$$

which generalizes the original Kuramoto model to normalized couplings on networks [34–39]. In this system, global phase coherence is quantified by the order parameter [37]

$$r(t) = \frac{1}{N} \left| \sum_{j=1}^N \exp[i\theta_j(t)] \right|, \quad (3)$$

with $\mathbf{i} = \sqrt{-1}$. In particular, near-synchronization configurations are characterized by fixing a threshold r^* close to 1 and defining the synchronization time as the first time t such that $r(t) = r^*$ [18, 40]. For random initial phases $\theta_i(0)$ uniformly distributed in $[0, 2\pi)$, synchronization times are treated as stochastic variables and analyzed statistically [18].

2.2. Definition of damage and functionality under edge perturbations

We take as reference the case $\Omega = \mathbf{A}$, i.e. original link capacities. To model damage on a specific link $a \rightarrow b$, we define Ω^* by setting $\Omega_{ab}^* = (1 - \beta)A_{ab}$ and $\Omega_{ij}^* = A_{ij}$ otherwise. Using equation (1), the couplings of a modified system are described by a matrix denoted as $\mathbf{W}^{(a,b)}(\beta)$ with elements $w_{i \rightarrow j}^{(a,b)}(\beta)$ given by [31, 32]

$$w_{i \rightarrow j}^{(a,b)}(\beta) \equiv \frac{\Omega_{ij}^*}{\sum_{l=1}^N \Omega_{il}^*} = \begin{cases} w_{i \rightarrow j}(\mathbf{A}) & \text{if } i \neq a, \\ \frac{A_{aj}}{k_a - \beta} & \text{if } i = a, j \neq b, \\ \frac{(1 - \beta)A_{ab}}{k_a - \beta} & \text{if } i = a, j = b, \end{cases} \quad (4)$$

so only the a -th row is modified. Here $k_a = \sum_{l=1}^N A_{al}$ and $\beta \rightarrow 0$ recovers $\mathbf{W}(\mathbf{A})$. The normalization redistributes the lost weight from $a \rightarrow b$ among the remaining links incident to node a [32].

Now, let τ_0 denote the synchronization time of the system with couplings $\mathbf{W}(\mathbf{A})$, and let $\tau(\beta)$ denote the synchronization time of the modified system defined by $\mathbf{W}^{(a,b)}(\beta)$, with identical random initial phases used in both dynamics. We define the functionality for this realization as [18]

$$\mathcal{F}_\beta \equiv \frac{\tau_0}{\tau(\beta)}, \quad (5)$$

and the global functionality as the ensemble average $\langle \mathcal{F}_\beta \rangle$. In particular, $\langle \mathcal{F}_{\beta=0} \rangle = 1$. The infinitesimal response to damage on $a \rightarrow b$ is [32]

$$\Lambda_{a,b}^{(\text{KM})} \equiv \left. \frac{d\langle \mathcal{F}_\beta \rangle}{d\beta} \right|_{\beta \rightarrow 0}. \quad (6)$$

With this definition, $\Lambda_{a,b}^{(\text{KM})} > 0$ indicates antifragility (damage speeds up synchronization), $\Lambda_{a,b}^{(\text{KM})} < 0$ indicates fragility, and $\Lambda_{a,b}^{(\text{KM})} = 0$ indicates no first-order effect. These measures apply only when both the reference and damaged dynamics reach the threshold r^* in finite time [32].

2.3. Linear approximation and spectral characterization

For phase differences $|\theta_j(t) - \theta_i(t)| \ll 1$, equation (2) admits the linear approximation [40–42]

$$\frac{d\theta_i(t)}{dt} \approx \sum_{j=1}^N w_{i \rightarrow j}(\Omega) [\theta_j(t) - \theta_i(t)] = - \sum_{j=1}^N (\delta_{ij} - w_{i \rightarrow j}(\Omega)) \theta_j(t), \quad (7)$$

where δ_{ij} is the Kronecker delta. Defining the normalized Laplacian $\hat{\mathcal{L}}(\Omega)$ with elements [43, 44]

$$\mathcal{L}_{ij}(\Omega) \equiv \delta_{ij} - w_{i \rightarrow j}(\Omega), \quad (8)$$

the linear dynamics becomes

$$\frac{d\theta_i(t)}{dt} = - \sum_{j=1}^N \mathcal{L}_{ij}(\Omega) \theta_j(t), \quad (9)$$

with solution

$$\theta_i(t) = \sum_{j=1}^N \left(e^{-t\hat{\mathcal{L}}(\Omega)} \right)_{ij} \theta_j(0). \quad (10)$$

Let $\{|\phi_\ell(\Omega)\rangle\}_{\ell=1}^N$ be right eigenvectors of $\hat{\mathcal{L}}(\Omega)$ with eigenvalues $\xi_\ell(\Omega)$, ordered by their real part $\Re\{\xi_\ell(\Omega)\}$. For connected networks, $\xi_1(\Omega) = 0$ and $0 < \Re\{\xi_2(\Omega)\} \leq \dots \leq \Re\{\xi_N(\Omega)\}$. Writing $Q_{i\ell}(\Omega) \equiv \langle i|\phi_\ell(\Omega)\rangle$ and $\Upsilon(t, \Omega) = \text{diag}(e^{-t\xi_1(\Omega)}, \dots, e^{-t\xi_N(\Omega)})$, we have

$$e^{-t\hat{\mathcal{L}}(\Omega)} = \mathbf{Q}(\Omega) \Upsilon(t, \Omega) \mathbf{Q}(\Omega)^{-1}. \quad (11)$$

Defining the left eigenvectors $\{|\bar{\phi}_\ell(\Omega)\rangle\}$ from $\mathbf{Q}(\Omega)^{-1}$, the phases $\theta_i(t)$ in equation (10) are

$$\theta_i(t) = \sum_{j=1}^N \sum_{\ell=1}^N e^{-t\xi_\ell(\Omega)} \langle i|\phi_\ell(\Omega)\rangle \langle \bar{\phi}_\ell(\Omega)|j\rangle \theta_j(0). \quad (12)$$

This shows that, in the linear regime, the relaxation is fully determined by the spectrum of $\hat{\mathcal{L}}(\Omega)$, connecting the dynamics to normalized diffusive transport on weighted networks [33].

Analogously to equation (5), we define the linear functionality using synchronization times from equation (9),

$$\mathcal{F}_\beta^{(\text{linear})} \equiv \frac{\tau_0^{(\text{linear})}}{\tau^{(\text{linear})}(\beta)}. \quad (13)$$

The average over multiple realizations yields the global functionality $\langle \mathcal{F}_\beta^{(\text{linear})} \rangle$. The corresponding infinitesimal modification then allows us to define

$$\Lambda_{a,b}^{(\text{linear})} \equiv \left. \frac{d \langle \mathcal{F}_\beta^{(\text{linear})} \rangle}{d\beta} \right|_{\beta \rightarrow 0}. \quad (14)$$

To reduce computational cost, a widely used spectral proxy is based on the second eigenvalue [45]. Defining $\tau_0^{(\xi_2)} \equiv 1/\Re\{\xi_2(\mathbf{A})\}$ and $\tau^{(\xi_2)}(\beta) \equiv 1/\Re\{\xi_2(\Omega^*)\}$, it is obtained [32]

$$\mathcal{F}_\beta^{(\xi_2)} \equiv \frac{\tau_0^{(\xi_2)}}{\tau^{(\xi_2)}(\beta)} = \frac{\Re\{\xi_2(\Omega^*)\}}{\xi_2(\mathbf{A})}, \quad (15)$$

where $\xi_2(\mathbf{A})$ is real for undirected connected networks. An important feature of equation (15) is that this quantity does not require averaging over initial conditions. The corresponding infinitesimal response is given by [32]

$$\Lambda_{a,b}^{(\xi_2)} \equiv \left. \frac{d\mathcal{F}_\beta^{(\xi_2)}}{d\beta} \right|_{\beta \rightarrow 0}. \quad (16)$$

3. Perturbative approach for $\Lambda_{a,b}^{(\xi_2)}$

In this section we develop an analytical framework to quantify how infinitesimal damage to a single link modifies the spectral properties governing synchronization. We first derive a general perturbative expression for the variation of the slowest relaxation mode, which provides a local and topology-dependent criterion to classify edges as fragile or antifragile. We then apply this framework to two representative classes of networks: the lollipop graph, as a deterministic system with extreme structural heterogeneity, and networks with community structure. Together, these results elucidate how antifragility emerges from the interplay between normalized coupling and modular organization.

3.1. Spectral variations induced by damage in a link

Let us now analyze the synchronization response to an infinitesimal reduction of the weight of a single edge (a, b) , using the damaged couplings defined by $\mathbf{W}^{(a,b)}(\beta)$ in equation (4), with $\beta \geq 0$ and $\beta \ll 1$. The corresponding normalized Laplacian is

$$\hat{\mathcal{L}}(\beta) = \mathbf{I} - \mathbf{W}^{(a,b)}(\beta), \quad (17)$$

where $\mathbf{I}$ is the $N \times N$ identity matrix. For $\beta \ll 1$, a first-order expansion yields

$$\mathbf{W}^{(a,b)}(\beta) \approx \mathbf{W} + \beta \left(\frac{\mathbf{E}^{(a,a)}\mathbf{W} - \mathbf{E}^{(a,b)}}{k_a} \right), \quad (18)$$

where $\mathbf{W} \equiv \mathbf{W}(\mathbf{A})$, $k_a = \sum_{j=1}^N A_{aj}$, and $(\mathbf{E}^{(a,b)})_{ij} \equiv \delta_{ia}\delta_{jb}$. Then, defining

$$\mathbf{\Gamma} \equiv \frac{\mathbf{E}^{(a,a)}\mathbf{W} - \mathbf{E}^{(a,b)}}{k_a}. \quad (19)$$

Equation (18) becomes

$$\mathbf{W}^{(a,b)}(\beta) \approx \mathbf{W} + \beta\mathbf{\Gamma}. \quad (20)$$

Substituting equation (20) into equation (17) we obtain

$$\hat{\mathcal{L}}(\beta) \approx \hat{\mathcal{L}}(0) - \beta\mathbf{\Gamma}, \quad (21)$$

with $\hat{\mathcal{L}}(0) = \mathbf{I} - \mathbf{W}$. Furthermore, let $\xi_2(0)$ denote the second eigenvalue of $\hat{\mathcal{L}}(0)$ and $\xi_2(\beta)$ the corresponding eigenvalue of $\hat{\mathcal{L}}(\beta)$. For $\beta \ll 1$,

$$\xi_2(\beta) = \xi_2(0) - \beta \Delta \xi_2(0) + o(\beta^2). \quad (22)$$

Using the definition of $\mathcal{F}_\beta^{(\xi_2)}$ in equation (15), the response to an infinitesimal modification of the link $a \rightarrow b$ is given by

$$\Lambda_{a,b}^{(\xi_2)} \equiv \left. \frac{d\mathcal{F}_\beta^{(\xi_2)}}{d\beta} \right|_{\beta \rightarrow 0} = \frac{1}{\xi_2(0)} \left. \frac{d\xi_2(\beta)}{d\beta} \right|_{\beta \rightarrow 0}. \quad (23)$$

Finally, combining equation (23) with the expansion in equation (22) we arrive at

$$\Lambda_{a,b}^{(\xi_2)} = -\frac{\Delta \xi_2(0)}{\xi_2(0)}. \quad (24)$$

Using first-order perturbation theory for the simple eigenvalue $\xi_2(\beta)$, and denoting by $|\phi_2(0)\rangle$ and $\langle \bar{\phi}_2(0)|$ the corresponding right and left eigenvectors of $\hat{\mathcal{L}}(0)$, we obtain

$$\Delta \xi_2(0) = \langle \bar{\phi}_2(0) | \mathbf{\Gamma} | \phi_2(0) \rangle. \quad (25)$$

To evaluate equation (25), we substitute the definition of $\mathbf{\Gamma}$ given in equation (19), obtaining

$$\langle \bar{\phi}_2(0) | \mathbf{\Gamma} | \phi_2(0) \rangle = \frac{1}{k_a} \left[\langle \bar{\phi}_2(0) | \mathbf{E}^{(a,a)}\mathbf{W} | \phi_2(0) \rangle - \langle \bar{\phi}_2(0) | \mathbf{E}^{(a,b)} | \phi_2(0) \rangle \right]. \quad (26)$$

However

$$\begin{aligned} \langle \bar{\phi}_2(0) | \mathbf{E}^{(a,a)}\mathbf{W} | \phi_2(0) \rangle &= \sum_{i=1}^N \sum_{m=1}^N \sum_{j=1}^N \langle \bar{\phi}_2(0) | i \rangle \left(\mathbf{E}^{(a,a)} \right)_{im} w_{m \rightarrow j} \langle j | \phi_2(0) \rangle \\ &= \sum_{i=1}^N \sum_{m=1}^N \sum_{j=1}^N \langle \bar{\phi}_2(0) | i \rangle \delta_{ia} \delta_{ma} w_{m \rightarrow j} \langle j | \phi_2(0) \rangle \\ &= \langle \bar{\phi}_2(0) | a \rangle \sum_{j=1}^N w_{a \rightarrow j} \langle j | \phi_2(0) \rangle. \end{aligned} \quad (27)$$

Similarly,

$$\begin{aligned}\langle \bar{\phi}_2(0) | \mathbf{E}^{(a,b)} | \phi_2(0) \rangle &= \sum_{i=1}^N \sum_{j=1}^N \langle \bar{\phi}_2(0) | i \rangle \left(\mathbf{E}^{(a,b)} \right)_{ij} \langle j | \phi_2(0) \rangle \\ &= \sum_{i=1}^N \sum_{j=1}^N \langle \bar{\phi}_2(0) | i \rangle \delta_{ia} \delta_{jb} \langle j | \phi_2(0) \rangle = \langle \bar{\phi}_2(0) | a \rangle \langle b | \phi_2(0) \rangle.\end{aligned}\quad (28)$$

Therefore

$$\Delta \xi_2(0) = \frac{\langle \bar{\phi}_2(0) | a \rangle}{k_a} \left[\sum_{j=1}^N w_{a \rightarrow j} \langle j | \phi_2(0) \rangle - \langle b | \phi_2(0) \rangle \right]. \quad (29)$$

Equation (29) shows that $\Delta \xi_2(0)$ can be either positive or negative. In particular, $\Delta \xi_2(0) < 0$ implies $\Lambda_{a,b}^{(\xi_2)} > 0$, meaning that the edge (a, b) is antifragile. This behavior is typically associated with networks displaying dense modules, such as cliques or communities.

3.2. Lollipop networks: intra-clique edges are antifragile

We now analyze a paradigmatic example of a class of networks with strong structural heterogeneity, namely the lollipop graph. The lollipop network consists of N nodes and is constructed by joining a complete graph (clique) of size n to a path graph of length $L = N - n$. The nodes in the clique are labeled $1, 2, \dots, n$, while the nodes in the path are labeled $n, n+1, \dots, N$, so that node n acts as an articulation point shared by both subgraphs. This topology naturally combines two distinct structural components: a densely connected module (the clique), and a sparsely connected structure (the path). As a result, the second eigenvalue $\xi_2(0)$ of the normalized Laplacian $\hat{\mathcal{L}}(0)$ is dominated by the weak coupling between the clique and the path, making the lollipop graph an ideal test case to investigate the effect of infinitesimal edge damage on synchronization and to illustrate the emergence of antifragility within modular structures.

We focus on an intra-clique edge (a, b) , with $a, b \in \{1, \dots, n-1\}$ and $a \neq b$, so that neither endpoint is the articulation node. In this case, all clique nodes $i \in \{1, \dots, n-1\}$ are structurally equivalent, hence the second right eigenvector of $\hat{\mathcal{L}}(0)$ takes a block form,

$$\langle i | \phi_2(0) \rangle = \begin{cases} c_1, & i \in \{1, \dots, n-1\}, \\ c_2, & i = n, \\ x_i, & i \in \{n+1, \dots, N\}, \end{cases} \quad (30)$$

where $\{x_i\}$ describes the variation along the path.

We now evaluate equation (29) for $a, b \in \{1, \dots, n-1\}$. Since a is a clique node distinct from n , its degree is $k_a = n-1$, and its outgoing weights satisfy $w_{a \rightarrow j} = 1/(n-1)$ for $j \in \{1, \dots, n\} \setminus \{a\}$ and $w_{a \rightarrow j} = 0$ otherwise. Using equation (29) and the block structure in equation (30), we obtain

$$\sum_{j=1}^N w_{a \rightarrow j} \langle j | \phi_2(0) \rangle = \frac{1}{n-1} \left[\sum_{j=1}^{n-1} c_1 (1 - \delta_{ja}) + c_2 \right] = \frac{(n-2)c_1 + c_2}{n-1}. \quad (31)$$

For an intra-clique neighbor $b \in \{1, \dots, n-1\}$, we have $\langle b | \phi_2(0) \rangle = c_1$, hence equation (29) yields

$$\Delta \xi_2(0) = \frac{\langle \bar{\phi}_2(0) | a \rangle}{k_a} \left[\frac{(n-2)c_1 + c_2}{n-1} - c_1 \right] = \frac{\langle \bar{\phi}_2(0) | a \rangle}{(n-1)^2} (c_2 - c_1). \quad (32)$$

In lollipop graphs, it is valid that $\langle \bar{\phi}_2(0) | a \rangle c_1 > 0$, since $\langle \bar{\phi}_2(0) | a \rangle$ and c_1 have the same sign. In addition, $\langle \bar{\phi}_2(0) | a \rangle c_1 > \langle \bar{\phi}_2(0) | a \rangle c_2$. Therefore,

$$\Delta \xi_2(0) < 0. \quad (33)$$

Finally, using $\Lambda_{a,b}^{(\xi_2)} = -\Delta \xi_2(0)/\xi_2(0)$ from equation (24), we conclude that

$$\Lambda_{a,b}^{(\xi_2)} = \frac{\langle \bar{\phi}_2(0) | a \rangle}{\xi_2(0)(n-1)^2} (c_1 - c_2) > 0, \quad (34)$$

so intra-clique edges (a, b) with $a, b \in \{1, \dots, n-1\}$ are antifragile in the lollipop network. This result highlights how strong structural heterogeneity promotes antifragile behavior in synchronization dynamics. In the lollipop graph, weakening an intra-clique edge enhances the effective coupling in the densely connected module, thereby accelerating global relaxation. More generally, this example illustrates how modular architectures can convert local damage into improved global performance.

3.3. Linear theory on networks with communities

We now specialize the linear approximation in equation (9) to networks with community structure, and derive analytic estimates for the slow relaxation mode and for the sign of $\Lambda_{a,b}^{(\xi_2)}$ under infinitesimal edge damage. Our approach combines coarse graining and standard perturbation theory.

3.3.1. Community coarse graining of the linear dynamics

Let the N nodes be partitioned into M communities $\{\mathcal{C}_\mu\}_{\mu=1}^M$, with sizes $n_\mu = |\mathcal{C}_\mu|$ and membership map $g(i) = \mu$ if $i \in \mathcal{C}_\mu$ [46]. We define community-averaged phases

$$\Theta_\mu(t) \equiv \frac{1}{n_\mu} \sum_{i \in \mathcal{C}_\mu} \theta_i(t), \quad (35)$$

and block edge counts

$$\omega_{\mu\nu} \equiv \sum_{i \in \mathcal{C}_\mu} \sum_{j \in \mathcal{C}_\nu} A_{ij}, \quad s_\mu \equiv \sum_{\nu=1}^M \omega_{\mu\nu}. \quad (36)$$

When the network is modular, intra-community dynamics occurs on a fast time scale, whereas inter-community exchange is slow. As a result, the dynamics can be approximated by an effective M -state coupling matrix obtained by aggregation. The corresponding community transition matrix is

$$P_{\mu\nu} \equiv \frac{\omega_{\mu\nu}}{s_\mu}, \quad (37)$$

that fulfills $\sum_{\nu=1}^M P_{\mu\nu} = 1$. In addition, the community Laplacian is defined as $\hat{\mathcal{L}}^{(c)} \equiv \mathbf{I}_M - \mathbf{P}$ where $\mathbf{I}_M$ is the $M \times M$ identity matrix, therefore $\hat{\mathcal{L}}^{(c)}$ has the elements

$$\left(\hat{\mathcal{L}}^{(c)}\right)_{\mu\nu} = \delta_{\mu\nu} - P_{\mu\nu}. \quad (38)$$

Projecting equation (9) onto the subspace spanned by community indicators yields the reduced dynamics

$$\frac{d\Theta_\mu(t)}{dt} \approx - \sum_{\nu=1}^M \left(\hat{\mathcal{L}}^{(c)}\right)_{\mu\nu} \Theta_\nu(t), \quad (39)$$

so that the slowest relaxation mode is controlled by $\xi_2^{(c)}$, the second eigenvalue of $\hat{\mathcal{L}}^{(c)}$. We then use

$$\xi_2(0) \approx \xi_2^{(c)}. \quad (40)$$

3.3.2. Two-community estimate for $\Lambda_{a,b}^{(\xi_2)}$

We now derive a simple estimate for $\Lambda_{a,b}^{(\xi_2)}$ in equation (24) for networks composed of two communities. For $M = 2$, equation (37) yields

$$\mathbf{P} = \begin{pmatrix} 1 - p_{12} & p_{12} \\ p_{21} & 1 - p_{21} \end{pmatrix}, \quad (41)$$

with $p_{12} \equiv \omega_{12}/s_1$ and $p_{21} \equiv \omega_{21}/s_2$. The eigenvalues of $\mathbf{P}$ are 1 and $1 - (p_{12} + p_{21})$, and therefore

$$\xi_2^{(c)} = p_{12} + p_{21}. \quad (42)$$

Strong modularity implies small p_{12}, p_{21} and therefore small $\xi_2(0)$, yielding slow linear synchronization.

We now connect the perturbative approach in equations (18)–(29) to a structure with two well-defined communities. In this case, in equation (29) the bracket compares the neighbor weighted average of the second right eigenvector around node a with its value at node b .

For two well-separated communities, the slow mode is approximately block-constant [47, 48]

$$\langle i|\phi_2(0)\rangle \approx \begin{cases} +c, & \text{if } g(i) = 1, \\ -d, & \text{if } g(i) = 2, \end{cases} \quad c, d > 0. \quad (43)$$

Then, assuming $a \in \mathcal{C}_1$, and denoting by k_a^{in} and k_a^{out} the number of neighbors of a in $\mathcal{C}_1$ and $\mathcal{C}_2$, respectively, with $k_a = k_a^{\text{in}} + k_a^{\text{out}}$, we obtain

$$\sum_{j=1}^N w_{a \rightarrow j}(\mathbf{A}) \langle j|\phi_2(0)\rangle \approx \frac{k_a^{\text{in}}}{k_a} c - \frac{k_a^{\text{out}}}{k_a} d. \quad (44)$$

Substituting equation (44) into equation (29) yields two representative cases: First, if the damaged edge is intra-community, $b \in \mathcal{C}_1$ and $\langle b|\phi_2(0)\rangle \approx c$, thus

$$\Delta\xi_2(0) \approx \frac{\langle \bar{\phi}_2(0)|a\rangle}{k_a} \left[\frac{k_a^{\text{in}}}{k_a} c - \frac{k_a^{\text{out}}}{k_a} d - c \right] = -\frac{\langle \bar{\phi}_2(0)|a\rangle}{k_a^2} k_a^{\text{out}} (c + d). \quad (45)$$

Secondly, if the damaged edge is inter-community, $b \in \mathcal{C}_2$ and $\langle b|\phi_2(0)\rangle \approx -d$, thus

$$\Delta\xi_2(0) \approx \frac{\langle \bar{\phi}_2(0)|a\rangle}{k_a} \left[\frac{k_a^{\text{in}}}{k_a} c - \frac{k_a^{\text{out}}}{k_a} d + d \right] = \frac{\langle \bar{\phi}_2(0)|a\rangle}{k_a^2} k_a^{\text{in}} (c + d). \quad (46)$$

In this manner, we obtain the estimate of $\Lambda_{a,b}^{(\xi_2)}$ for $a \in \mathcal{C}_1$

$$\Lambda_{a,b}^{(\xi_2)} \approx \begin{cases} \frac{\langle \bar{\phi}_2(0)|a\rangle}{\xi_2(0)} \frac{k_a^{\text{out}} (c + d)}{k_a^2}, & b \in \mathcal{C}_1, \\ -\frac{\langle \bar{\phi}_2(0)|a\rangle}{\xi_2(0)} \frac{k_a^{\text{in}} (c + d)}{k_a^2}, & b \in \mathcal{C}_2. \end{cases} \quad (47)$$

Therefore, under the conventional orientation where $\langle \bar{\phi}_2(0)|a\rangle > 0$ for $a \in \mathcal{C}_1$, equations (45) and (46) imply

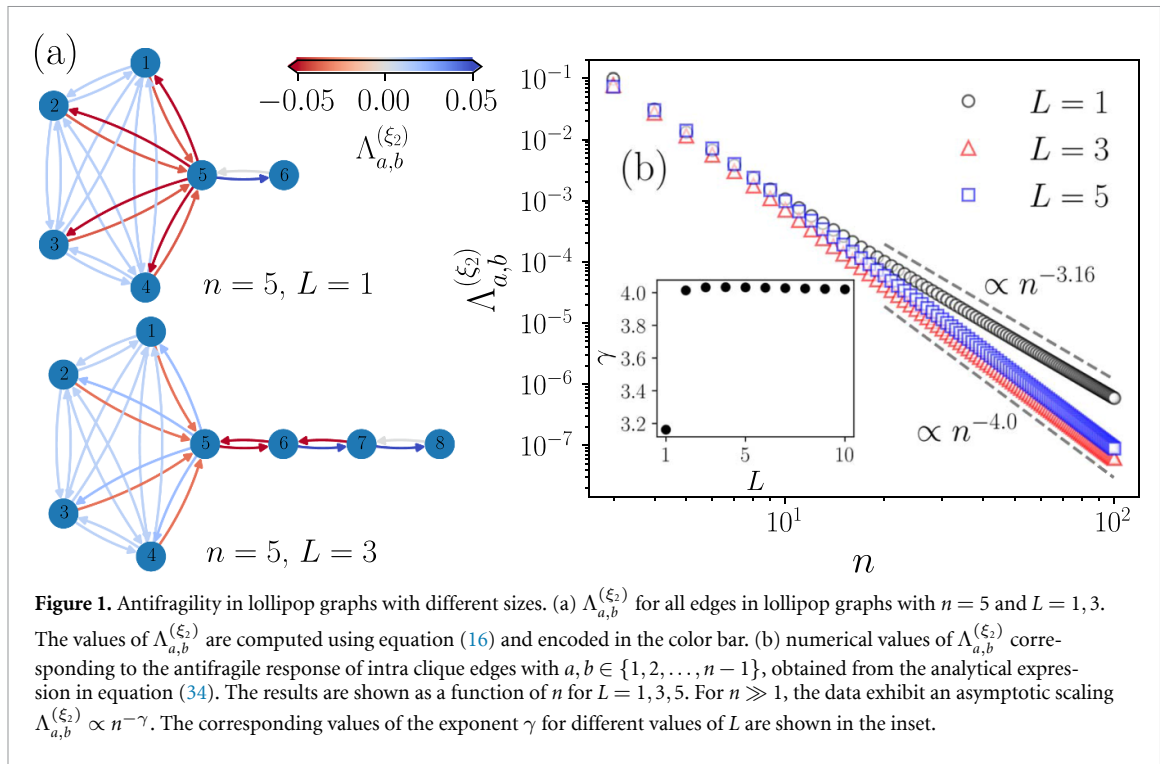
$$\Lambda_{a,b}^{(\xi_2)} > 0 \quad \text{for intra-community damage,} \quad (48)$$

$$\Lambda_{a,b}^{(\xi_2)} < 0 \quad \text{for inter-community damage.} \quad (49)$$

The analytical predictions in equations (48) and (49) show that edges located within dense communities often display antifragile behavior, whereas edges connecting different communities tend to produce fragile responses. This result provides an analytical mechanism for the emergence of antifragile edges inside dense modules. Due to the normalization of the couplings, weakening an intra-community link effectively increases the relative weight of the remaining links from node a , thereby increasing $\xi_2(\beta)$ and reducing the proxy $\tau^{(\xi_2)}(\beta) = 1/\Re\{\xi_2(\beta)\}$.

Our findings presented in this section show that the perturbative expression in equation (29) provides a transparent analytical criterion to understand antifragility in networks with community structure. The first-order correction $\Delta\xi_2(0)$ compares the weighted neighbor average of the slow mode around node a with its value at the damaged neighbor b . In modular networks, where the second right eigenvector is approximately block-constant across communities, this comparison acquires a clear structural meaning. For two well-separated communities, the slow mode takes opposite signs on each module, and the sign of $\Delta\xi_2(0)$ is controlled by whether the damaged edge (a, b) lies within a community or connects different communities. Weakening an intra-community edge, due to normalization, redistributes weight toward inter-community connections, enhancing global synchronization. As a result, $\Delta\xi_2(0) < 0$ and $\Lambda_{a,b}^{(\xi_2)} > 0$, indicating antifragility. In contrast, damaging an inter-community edge suppresses the already weak coupling between modules, leading to $\Delta\xi_2(0) > 0$ and a fragile response.

This interpretation follows from a perturbative treatment of the damage process. Accordingly, our analytical results apply to the regime of infinitesimal or very weak damage, where the first-order expansion remains valid and allows us to identify antifragile links analytically. While stronger damage would generally require going beyond the perturbative framework, the present analysis already captures the structural mechanism by which normalized coupling and modular topology determine whether a local perturbation leads to antifragile or fragile responses, with the slowest relaxation mode acting as a probe of how local damage reshapes global synchronization.



4. Numerical results

In this section, we present numerical results that validate and complement the analytical framework developed in section 3. Our goal is to assess how infinitesimal structural damage affects synchronization dynamics across different classes of networks. To this end, we consider networks with progressively increasing structural complexity, chosen to reflect and test the structural assumptions underlying the perturbative analysis. We begin with lollipop graphs, which provide a controlled and analytically tractable setting combining a dense clique and a sparse tail, thereby exhibiting extreme structural heterogeneity. This simple topology allows us to clearly illustrate the mechanisms predicted by the perturbative theory. We then move to random networks with community structure, first considering the minimal case of two communities and subsequently more complex modular architectures, in order to test whether the predicted effects persist in more realistic modular systems. Finally, we examine real-world networks drawn from different domains, including social, biological, and neuroscientific systems, all of which are known to exhibit pronounced modular organization. Together, these numerical experiments demonstrate that antifragility is a robust and generic consequence of modular topology.

4.1. Antifragility in lollipop graphs

We first investigate antifragility in lollipop graphs, which serve as a paradigmatic example of a highly heterogeneous network. A lollipop graph consists of a densely connected clique attached to a one-dimensional chain, a structure known to produce strong spectral separation and localization of the slowest Laplacian eigenmodes. These features make lollipop graphs an ideal testbed for probing the sensitivity of the second eigenvalue to localized perturbations. In this subsection, we focus on the spectral response measure $\Lambda_{a,b}^{(\xi_2)}$, which quantifies how an infinitesimal reduction of the weight of a single edge (a, b) modifies the slowest relaxation rate governing synchronization.

In figure 1(a) we report the values of $\Lambda_{a,b}^{(\xi_2)}$ for all edges in lollipop graphs with clique size $n = 5$ and chain lengths $L = 1$ and $L = 3$. The values are computed using equation (16), with the derivative evaluated numerically using an increment $\Delta\beta = 10^{-6}$. The results show a clear structural differentiation: edges connecting nodes $a, b \in \{1, 2, \dots, 4\}$ within the clique display an identical positive response, indicating antifragility, while several edges along the chain and those connecting the clique to the chain have negative values. Moreover, increasing the chain length enhances this contrast, reflecting a growing separation between the effects of intra clique perturbations and modifications of the couplings along the tail.

In figure 1(b) we focus on intra clique edges with $a, b \in \{1, 2, \dots, n-1\}$ and quantify their antifragile response as a function of the clique size n . The values of $\Lambda_{a,b}^{(\xi_2)}$ are obtained from the analytical expression in equation (34) and are shown for $L = 1, 3, 5$, the results coincide with the values obtained numerically using equation (16). Furthermore, for all considered values of L , the response decreases monotonically with n , indicating that smaller cliques exhibit a stronger antifragile effect. In the asymptotic regime $n \gg 1$, the results follow a power-law scaling $\Lambda_{a,b}^{(\xi_2)} \propto n^{-\gamma}$, with an exponent γ that depends on the chain length. The inset reports the corresponding values of γ , showing that for longer chains the exponent stabilizes around $\gamma \approx 4$.

Taken together, the results in figure 1 numerically demonstrate the validity of equation (34), showing that antifragility in lollipop graphs is strongly shaped by the interplay between local connectivity and global network structure. Highly connected cliques exhibit an antifragile response to local perturbations. The observed scaling behavior further highlights how antifragility diminishes with increasing clique size and becomes increasingly constrained by the size of the network. These findings provide a clear illustration of how modular heterogeneity governs the emergence of antifragile responses in synchronization in networked systems, and they serve as a benchmark for understanding similar effects in more general heterogeneous networks.

4.2. Antifragility in networks with two communities

Now, we numerically investigate antifragility in random networks with a prescribed two-community structure. The networks are generated using a stochastic block model with two equally sized communities $\mathcal{C}_1$ and $\mathcal{C}_2$, each containing n nodes. Pairs of nodes within the same community are connected independently with probability $1 - p$, while connections between nodes belonging to different communities are established with probability p . For small values of p , this construction yields networks with strong modular organization, characterized by dense intra-community connectivity and sparse inter-community coupling. As discussed in section 3.3, such networks exhibit a clear separation of time scales, with fast synchronization inside communities and slow relaxation associated with inter-community exchange, making them a natural benchmark for testing the perturbative predictions derived for $\Lambda_{a,b}^{(\xi_2)}$.

To quantify the typical response of different classes of edges, we introduce ensemble-averaged measures of antifragility. Let $\mathcal{E}_{\text{intra}}$ denote the set of edges whose endpoints belong to the same community, and $\mathcal{E}_{\text{inter}}$ the set of edges connecting nodes in different communities. For a given network realization, we define the average intra-community response as

$$\bar{\Lambda}_{\text{intra}}^{(\xi_2)} \equiv \frac{1}{|\mathcal{E}_{\text{intra}}|} \sum_{(a,b) \in \mathcal{E}_{\text{intra}}} \Lambda_{a,b}^{(\xi_2)}, \quad (50)$$

where $\Lambda_{a,b}^{(\xi_2)}$ is computed from equation (16) for each edge (a, b) . Since the networks are random, we further average this quantity over many independent realizations of the stochastic block model, and denote the resulting ensemble average by $\langle \bar{\Lambda}_{\text{intra}}^{(\xi_2)} \rangle$.

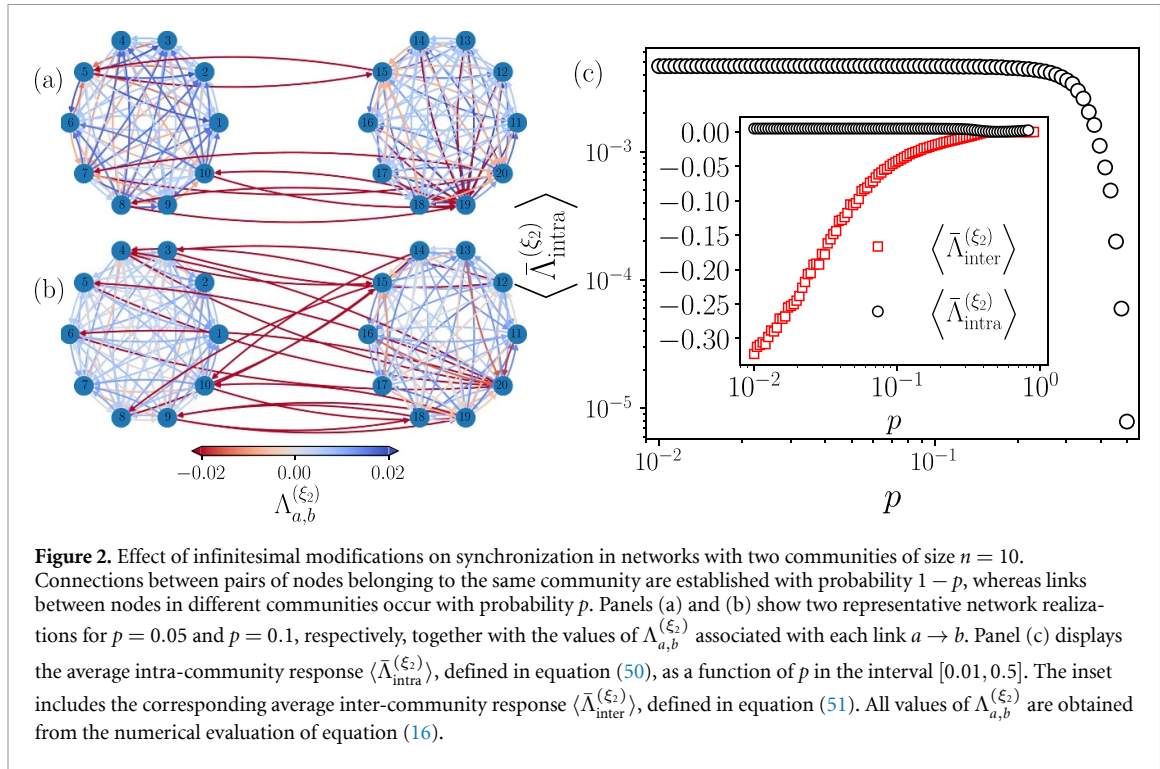
Analogously, for edges connecting different communities, we define

$$\bar{\Lambda}_{\text{inter}}^{(\xi_2)} \equiv \frac{1}{|\mathcal{E}_{\text{inter}}|} \sum_{(a,b) \in \mathcal{E}_{\text{inter}}} \Lambda_{a,b}^{(\xi_2)}, \quad (51)$$

and denote its ensemble average by $\langle \bar{\Lambda}_{\text{inter}}^{(\xi_2)} \rangle$.

The ensemble averages $\langle \bar{\Lambda}_{\text{intra}}^{(\xi_2)} \rangle$ and $\langle \bar{\Lambda}_{\text{inter}}^{(\xi_2)} \rangle$ enable a direct comparison of the typical effect of infinitesimal damage on intra and inter-community links, and allow us to assess how antifragility emerges as a function of the inter-community coupling probability p .

Figure 2 illustrates the effect of infinitesimal edge modifications on synchronization in networks with two communities of equal size $n = 10$. Panels 2(a) and (b) show two representative realizations of the stochastic block model for inter-community connection probabilities $p = 0.05$ and $p = 0.1$, respectively. In each network, edges are color-coded according to the corresponding values of $\Lambda_{a,b}^{(\xi_2)}$, allowing a direct visualization of antifragile and fragile links. For small values of p , when the modular structure is strong, intra-community edges predominantly display positive values of $\Lambda_{a,b}^{(\xi_2)}$, indicating antifragile behavior, whereas inter-community edges are mostly fragile. As p increases and the distinction between communities becomes less pronounced, this contrast progressively weakens. The results in figure 2(c) quantify this behavior by showing the ensemble-averaged intra-community response $\langle \bar{\Lambda}_{\text{intra}}^{(\xi_2)} \rangle$, defined in equation (50), as a function of the inter-community coupling probability p in the range $0.01 \leq p \leq 0.5$. Each data point

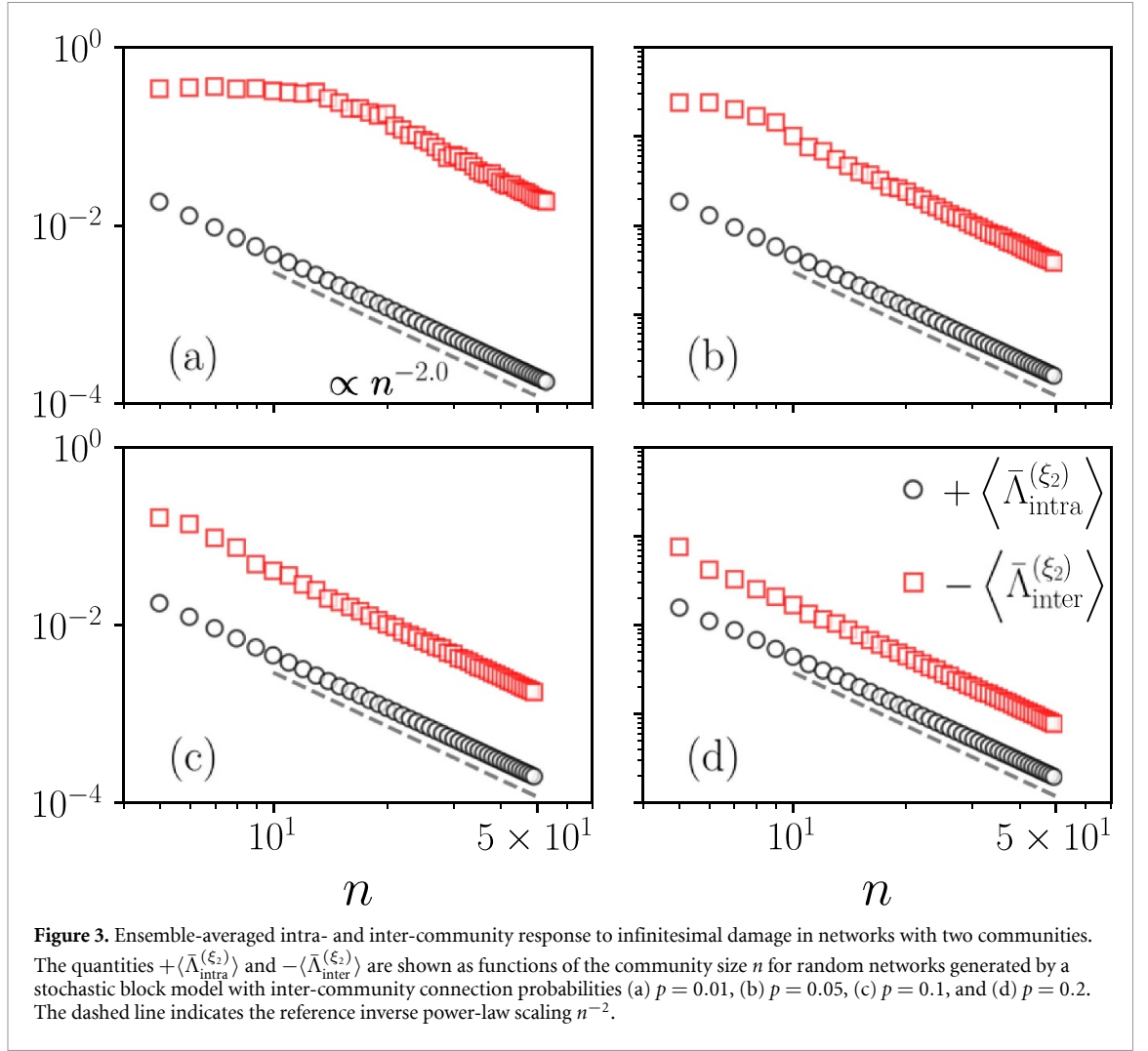


is obtained by averaging over 10^3 independent realizations of the network. The inset reports the corresponding ensemble-averaged inter-community response $\langle \bar{\Lambda}_{\text{inter}}^{(\xi_2)} \rangle$, defined in equation (51). For small p , $\langle \bar{\Lambda}_{\text{intra}}^{(\xi_2)} \rangle$ remains positive and constant, confirming that infinitesimal damage to intra-community links typically enhances global synchronization, in agreement with the analytical predictions of section 3.3. In contrast, $\langle \bar{\Lambda}_{\text{inter}}^{(\xi_2)} \rangle$ is negative over the same range, reflecting the fragile role of inter-community connections. As p increases, both averages approach zero, signaling the loss of antifragile effects.

On the other hand, figure 3 summarizes how the typical response of intra- and inter-community edges depends on the community size n in networks composed of two communities. For all explored values of the inter-community connection probability p , the ensemble-averaged intra-community response $+\langle \bar{\Lambda}_{\text{intra}}^{(\xi_2)} \rangle$ remains positive, confirming that infinitesimal damage to intra-community links systematically enhances global synchronization. In contrast, $-\langle \bar{\Lambda}_{\text{inter}}^{(\xi_2)} \rangle$ is also positive, indicating that weakening inter-community links produces a fragile effect on synchronization. The reported values are obtained from the numerical evaluation of equation (16) together with equations (50) and (51), and are in excellent agreement with the results computed using the perturbative expression in equation (29).

The numerical results in figures 3(a)–(d) reveal that the quantities $+\langle \bar{\Lambda}_{\text{intra}}^{(\xi_2)} \rangle$ and $-\langle \bar{\Lambda}_{\text{inter}}^{(\xi_2)} \rangle$ decrease monotonically with increasing community size n , and are well described by an inverse power-law scaling proportional to n^{-2} in the asymptotic regime $n \gg 1$, indicated by the dashed reference line. This scaling is robust across different levels of modularity, ranging from strongly modular networks ($p = 0.01$) to more weakly modular ones ($p = 0.2$). The observed decay reflects the fact that, as communities grow, the relative impact of perturbing a single edge on the slowest relaxation mode becomes increasingly diluted. Consequently, antifragility is strongest in small, well-defined modules, whereas in large communities both antifragile and fragile effects become asymptotically weak, even though their qualitative distinction persists.

Our findings for random networks with two communities confirm the perturbative predictions developed in section 3.3. Infinitesimal damage to intra-community links is typically antifragile, whereas damage to inter-community links is fragile, reflecting their opposite influence on the slowest relaxation mode. Both effects decrease systematically with increasing community size, following a robust inverse power-law scaling, which indicates that single-edge perturbations become progressively less relevant in large modules. Altogether, these results demonstrate that antifragility emerges naturally from modular organization and spectral separation, and that the two-community network constitutes a minimal and transparent setting for understanding this phenomenon.



4.3. Random networks with communities

The results obtained so far demonstrate that antifragility naturally emerges in networks with random community structure. We now extend this analysis to more realistic and heterogeneous topologies by considering networks with multiple communities generated using the Lancichinetti–Fortunato–Radicchi (LFR) benchmark model [49]. LFR networks are widely used to mimic salient features of real-world systems, including heterogeneous degree distributions, broad community size distributions, and tunable levels of inter-community mixing.

As a natural extension of the edge-level response $\Lambda_{a,b}^{(\xi_2)}$, we introduce a community-level measure of antifragility. For a given community $\mathcal{C}_\mu$, we define

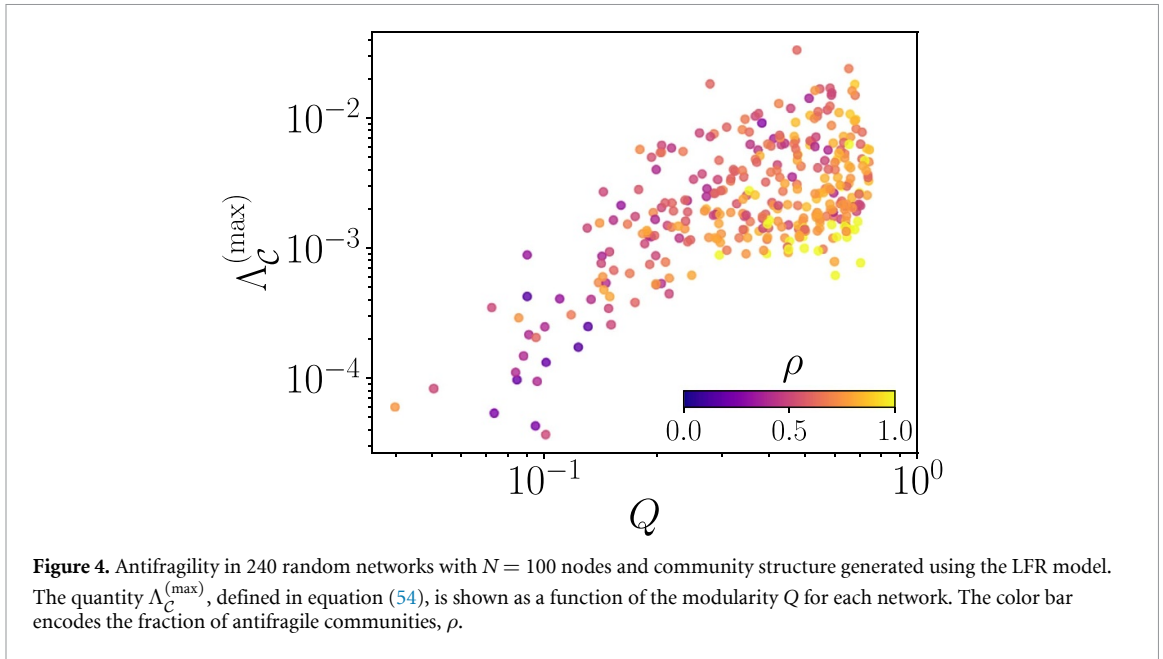
$$\bar{\Lambda}_{\mathcal{C}_\mu} \equiv \frac{1}{|\mathcal{E}_\mu|} \sum_{(a,b) \in \mathcal{E}_\mu} \Lambda_{a,b}^{(\xi_2)}, \quad (52)$$

where $\mathcal{E}_\mu \subset \mathcal{E}$ denotes the set of edges with both endpoints in $\mathcal{C}_\mu$, and $|\mathcal{E}_\mu|$ is the corresponding number of edges. The quantity $\Lambda_{a,b}^{(\xi_2)}$ is computed for each edge (a,b) using equation (16). The average $\bar{\Lambda}_{\mathcal{C}_\mu}$ thus quantifies the typical response of community $\mathcal{C}_\mu$ to infinitesimal internal damage; in this manner, $\bar{\Lambda}_{\mathcal{C}_\mu} > 0$ indicates antifragile behavior at the community-level, whereas $\bar{\Lambda}_{\mathcal{C}_\mu} < 0$ signals fragility.

In addition, to characterize the strength of the community organization, we use the modularity Q , which for an undirected network partitioned into M communities is defined as

$$Q \equiv \sum_{\mu=1}^M \left[\frac{|\mathcal{E}_\mu|}{|\mathcal{E}|} - \left(\frac{k_{\mathcal{C}_\mu}}{2|\mathcal{E}|} \right)^2 \right], \quad (53)$$

where $k_{\mathcal{C}_\mu} = \sum_{i \in \mathcal{C}_\mu} k_i$ is the sum of degrees of the nodes within community $\mathcal{C}_\mu$. The modularity Q quantifies the excess of intra-community connections relative to a randomized null model with the same



degree sequence [50–52], and therefore provides a natural scalar measure of the quality and strength of the community structure.

In order to explore the emergence of antifragility in LFR networks, we generate an ensemble of 240 realizations with $N = 100$ nodes, spanning a broad range of modularity values $0.238 \leq Q \leq 0.742$. This variability is obtained by tuning the mixing parameter η , which controls the fraction of edges connecting nodes to communities other than their own, from $\eta = 0.05$, corresponding to strongly modular networks, to $\eta = 0.5$, corresponding to weakly modular ones [49]. To further increase structural heterogeneity, we consider degree distribution exponents equal to 2 and 3, together with community size distribution exponents equal to 1 and 2. For each realization, the LFR model generates both the network topology and its associated community partition. Using this partition, the modularity Q is computed with the NetworkX package (version 3.4.2) [53].

Figure 4 presents the analysis of antifragility at the community-level for an ensemble of 240 networks with modular structure. For each network, we first compute the edge-level response $\Lambda_{a,b}^{(\xi_2)}$ using equation (16). Using the community partition provided by the LFR model, these edge-level responses are then aggregated to obtain the community-averaged antifragility $\bar{\Lambda}_{C_\mu}$ defined in equation (52). From these values, we extract the maximal community response

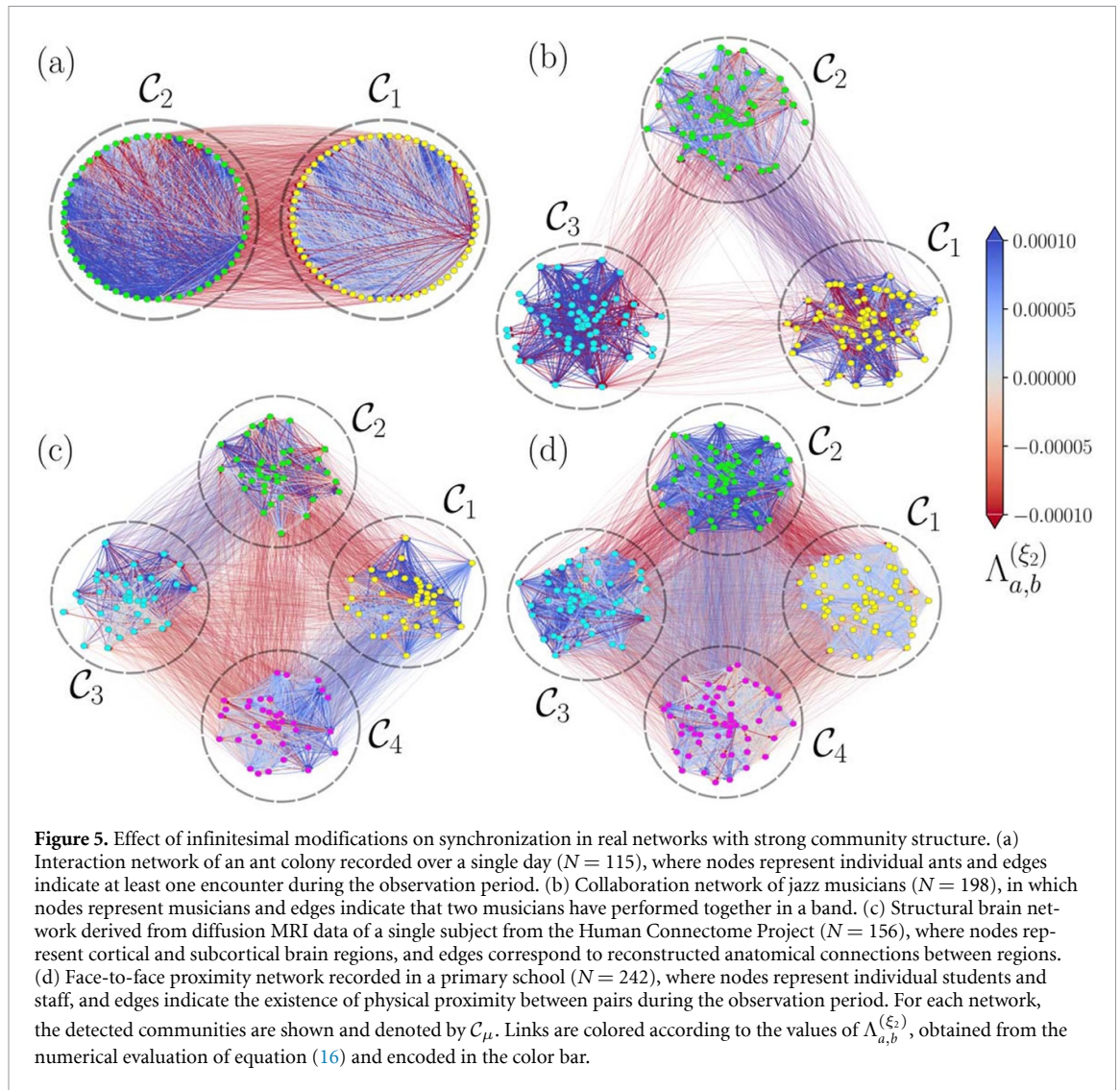
$$\Lambda_C^{(\max)} \equiv \max\{\bar{\Lambda}_{C_\mu} : \mu = 1, \dots, M\}, \quad (54)$$

which quantifies the strongest antifragile behavior observed among all communities within a given network. Figure 4 displays $\Lambda_C^{(\max)}$ as a function of the modularity Q for each realization. In addition, for every network we compute the fraction ρ of communities satisfying $\bar{\Lambda}_{C_\mu} > 0$, indicating antifragile behavior at the community scale. The corresponding values of ρ are encoded in the color bar.

Two robust trends emerge from these results. First, every network in the ensemble exhibits at least one antifragile community, showing that antifragility is a generic feature of modular networks rather than a rare or fine-tuned outcome. Second, the fraction of antifragile communities ρ typically exceeds 50%, indicating that antifragility often involves a substantial portion of the mesoscopic organization. In addition, the magnitude of the strongest community-level response, $\Lambda_C^{(\max)}$, increases systematically with the modularity Q , demonstrating that stronger community structure enhances the ability of networks to convert local damage into improved global synchronization. These findings are in full agreement with the perturbative mechanism identified in section 3 and confirm that the proposed framework captures antifragile behavior in heterogeneous networks beyond idealized or minimally structured cases.

4.4. Real networks with communities

We now extend our analysis to real-world networks with empirically observed community structure, in order to assess whether the antifragile mechanisms identified in synthetic models persist in realistic settings. We analyze four networks drawn from different domains, including social, biological, and neuroscientific systems, all of which are known to exhibit pronounced modular organization. These networks



are shown in figure 5. In all cases, we restrict the analysis to the largest connected component (LCC) of each network, ensuring that the normalized Laplacian has a unique zero eigenvalue and that the second eigenvalue ξ_2 is well defined.

Figure 5(a) shows an interaction network of an ant colony recorded over a single day [54]. Nodes represent individual ants ($N = 115$), and an undirected edge indicates that two ants were observed interacting at least once during the recording period. This system exhibits strong spatial and behavioral organization. After restricting the analysis to the LCC, community detection reveals two well-defined communities, corresponding to functionally coherent groups of ants within the colony.

Figure 5(b) displays a collaboration network of jazz musicians [55], where nodes represent musicians ($N = 198$) and edges indicate that two musicians have performed together in the same band. This network constitutes a paradigmatic example of social modularity, with communities shaped by stylistic affinities, historical collaborations, and shared band membership. The resulting topology is organized into three main groups of nodes, each characterized by dense intra-community connectivity and comparatively sparse inter-community coupling.

In figure 5(c), we consider a structural brain network reconstructed from diffusion MRI data of a single subject from the Human Connectome Project [56]. Nodes ($N = 156$) correspond to cortical and subcortical regions defined by the L2018 atlas at intermediate resolution, while edges represent reconstructed anatomical connections between pairs of regions. After restricting the analysis to the LCC, the network exhibits a pronounced modular organization consisting of four well-defined communities.

Figure 5(d) shows a face-to-face proximity network recorded in a primary school [57]. Nodes represent students and staff members ($N = 242$), and edges indicate physical proximity events detected during the observation period. After restricting the analysis to the LCC, the network exhibits a clear modular organization with four communities, primarily shaped by classroom structure and scheduled activities.

For each network shown in figure 5, communities C_μ are identified using modularity-based methods, and edges are colored according to the corresponding values of $\Lambda_{a,b}^{(\xi_2)}$, obtained from the numerical evaluation of equation (16). Across all systems, intra-community edges predominantly exhibit positive values of $\Lambda_{a,b}^{(\xi_2)}$. Accordingly, for the networks analyzed, all detected communities satisfy $\bar{\Lambda}_{C_\mu} > 0$, indicating antifragile behavior at the community-level. In contrast, inter-community edges are more likely to display negative values of $\Lambda_{a,b}^{(\xi_2)}$, reflecting fragile behavior. We note, however, that antifragility may also arise in specific inter-community connections. An illustrative example is provided by the links connecting communities C_1 and C_4 in the structural brain network shown in figure 5(c).

Our findings demonstrate that the conversion of local damage into enhanced global synchronization is not limited to idealized models, but also occurs in real systems with heterogeneous structure and empirically grounded community organization. The persistence of antifragile edges across diverse domains underscores modular topology as a general structural mechanism governing the response of complex dynamical networks to infinitesimal perturbations.

5. Conclusions

In this work, we have developed an analytical and numerical framework to characterize antifragility in synchronization dynamics on networks, focusing on the effect of structural damage defined as a reduction in the weight of an edge connecting two nodes. By studying identical Kuramoto oscillators with normalized couplings, we showed that the response of synchronization times to local damage can be naturally quantified through a spectral measure based on the second eigenvalue of the normalized Laplacian. This approach provides a transparent connection between local structural perturbations and global dynamical performance. Using first-order perturbation theory, we derived a general expression for the infinitesimal response $\Lambda_{a,b}^{(\xi_2)}$ and demonstrated that its sign is controlled by the structure of the slowest relaxation mode. This result reveals that networks with strong structural heterogeneity and modular organization are therefore natural candidates to exhibit antifragile behavior. These results provide a spectral explanation for the antifragile responses observed numerically in synchronization dynamics in [32].

We illustrated these mechanisms analytically and numerically in lollipop graphs and in networks with community structure. In both settings, intra-community edges were shown to be typically antifragile, while inter-community edges were fragile, reflecting their opposite roles in response to damage. Numerical simulations confirmed the analytical predictions and revealed robust scaling laws, showing that antifragile effects decay algebraically with increasing community size. This establishes that antifragility is strongest in small, well-defined modules and becomes progressively weaker in large communities, although the qualitative distinction between intra- and inter-community links persists.

Extending the analysis to networks with many communities generated by the LFR algorithm and to several real-world networks from different domains, including social, biological, and neuroscientific systems, we found that antifragility is a widespread and robust phenomenon. In all cases studied, at least one community exhibited an antifragile response, with edges located inside dense modules tending to enhance global synchronization when weakened, while edges bridging different communities typically produced fragile responses. Moreover, the prevalence and magnitude of antifragility increased systematically with the strength of modular organization, as quantified by the community structure of the networks. These observations were consistently reproduced across synthetic and empirical networks, indicating that the mechanism identified in this work is not an artifact of idealized models but rather a structural consequence of modular topology in complex systems.

Overall, our results clarify how modular topology and spectral separation enable the conversion of local damage into improved global synchronization. Beyond synchronization, the perturbative and spectral approach developed here provides a general framework to investigate antifragility in a broad class of dynamical processes on networks. From a practical perspective, these findings suggest design principles for engineered systems in which controlled weakening of specific interactions can be used to enhance performance, and offer new insight into how complex systems may exploit structural heterogeneity to benefit from stress or damage at different scales.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Author contributions

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